INVERTIS UNIVERSITY BAREILLY
Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

BEC-101- Emerging Domain In Electronics Engineering

L T P C
3 0 0 3

Unit	Topics
I	**Introduction:** Semiconductors, Extrinsic & Intrinsic type, doping, p-n junction diode **Semiconductor Diode**: Depletion layer, V-I characteristics, ideal and practical Diodes, Diode Equivalent Circuits, Zener Diodes breakdown mechanism (Zener and avalanche) **Diode Application:** Diode Configuration, Half and Full Wave rectification, Clippers,Clampers, Zener diode as shunt regulator, Voltage-Multiplier Circuits
II	**Bipolar Junction Transistor:** Transistor Construction, Operation, Amplification action. Common Base, Common Emitter, Common Collector Configuration **Field Effect Transistor:** Construction and Characteristic of JFETs. TransferCharacteristic. MOSFET (MOS) (Depletion and Enhancement) Type, Transfer Characteristic.
III	**Operational Amplifiers:** Introduction, Op-Amp Basic, Practical Op-Amp Circuits(Inverting Amplifier, Non-inverting Amplifier, Unit Follower, Summing Amplifier,Integrator, Differentiator). Differential and Common-Mode Operation, Comparators.
IV	**Digital Electronics:** Number system & representation. Introduction of Basic and Universal Gates, using Boolean algebra simplification of Boolean function. K Map.
V	**CRO as Measurement Instrument:** CRO (its working with block diagram), measurement of voltage, current, and frequency using CRO. **Introduction of IoT System,** Components of IoT system, Basic Introduction to Sensors

Text Books:

1. Robert L. Boylestand / Louis Nashelsky "Electronic Devices and Circuit Theory", PearsonEducation.
2. H S Kalsi, "Electronic Instrumentation", McGraw Publication
3. George Kennedy, "Electronic Communication Systems", McGraw Publication
4. David A. Bell, "Electronic Devices and Circuits", Oxford University Press.
5. Jacob Millman, C.C. Halkias, Staya brataJit, "Electronic Devices and Circuits", McGraw Hill
6. David A. Bell, Electronic Instrumentation and Measurements, Latest Edition, Oxford University Press India

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Understand the concept of PN Junction and devices.
2. Understand the concept of BJT, FET and MOFET.
3. Understand the concept of Operational amplifier
4. Understand the concept of measurement instrument.
5. Understand the working principle of different type of sensor and their uses.
6. Understand the concept of IoT system & understand the component of IoT system